SDG 11: Sustainable Cities and Communities

Inclusive Urban Resilience through Municipal Access

Introduction

Sustainable Development Goal 11 (SDG 11) aims to make cities and human settlements inclusive, safe, resilient, and sustainable. In Sri Lanka, this goal is closely linked to the everyday systems that shape urban life: flood preparedness, drainage, transport, waste collection, housing-related notices, public spaces, and municipal service communication.

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This use case focuses on city-level communication barriers rather than water quality, energy access, job creation, digital infrastructure, or inequality as separate themes. Sri Lankan cities face rapid urbanization, transport congestion, weak drainage, growing waste-management pressures, and uneven access to clear public information during emergencies and service disruptions.

In Colombo, research shows that urban expansion, reduced green spaces, loss of natural water-retention zones, and inadequate drainage infrastructure have increased surface runoff and flood vulnerability, particularly in areas with informal settlements. Wider urban infrastructure studies also identify transport, utilities, housing, green spaces, water, electricity, and waste management as core concerns for urban residents.

This report explores how VosynCore can support SDG 11 by strengthening multilingual municipal communication and urban resilience. Through the narrative of Amara--a resident in a flood-prone urban neighbourhood--we illustrate how accessible language support can help households act on city-level information.

Modelled projections suggest that this municipal communication architecture could improve flood-alert comprehension by up to 45%, increase correct local service reporting by 50%, improve waste-instruction understanding across pilot communities, and strengthen public participation in neighbourhood planning. By embedding multilingual voice guidance into municipal service communication, VosynCore provides a scalable pathway toward SDG 11 targets on safe transport, participatory planning, disaster-risk reduction, waste management, and resilient cities.

Use Context: Urban Resilience, Mobility & Municipal Communication

Sustainable cities depend not only on physical infrastructure, but also on whether residents receive clear, timely, and usable information. A city may have drainage channels, transport routes, waste collection schedules, and disaster response plans, but those systems are less effective when residents cannot understand warnings, reporting channels, or service instructions. In this sense, municipal communication is part of urban infrastructure itself.

A central challenge for SDG 11 is that urban residents often interact with multiple systems at once. A heavy rainfall event may trigger flood warnings, traffic diversions, school closures, waste delays, and drainage complaints. If each update arrives through a different channel or in formal technical language, residents may not know which action to take first or where to seek reliable help.

Urban flooding is one of the clearest SDG 11 concerns for Sri Lanka. Colombo has experienced land use and land cover changes linked to rapid urbanization, reduced green areas, and loss of water-retention spaces. These changes, combined with drainage limitations and informal settlement vulnerability, have contributed to recurring urban flood risk and demonstrate why city-level resilience communication matters.

> **Example: A resident may receive a flood warning, waste-collection notice, or transport diversion update through an official website, notice board, SMS, or social media post, but may not understand what action is required, which road to avoid, where to report a blocked drain, or how to prepare a household before water levels rise.**

Urban infrastructure challenges also extend to mobility and public services. Research on Sri Lankan urban areas highlights traffic congestion, insufficient housing, ineffective utilities, environmental damage, and the need for better future infrastructure planning. These are not only engineering issues; they affect how residents move, access services, and participate in local decisions.

Traditional responses such as printed notices, English-heavy websites, office counters, or one-time awareness sessions are useful but insufficient for fast-moving urban challenges. Residents need accessible, multilingual, voice-enabled support that can explain local instructions in practical language and guide them toward appropriate action.

Use Case Narrative: Amara's Journey from Confusion to Urban Resilience

To illustrate the human stakes and potential solutions, consider the story of Amara, a 41-year-old resident living in a flood-prone neighbourhood within the Colombo urban region.

Amara lives with her two children and her elderly father. She works part-time from home and depends on buses, local shops, school notices, and municipal services to keep her household running. During the rainy season, her neighbourhood often faces waterlogging, blocked drains, transport delays, and confusing updates from different agencies.

In theory, city systems are designed to protect residents like Amara. Flood alerts are issued, waste schedules are updated, roads are closed when necessary, and municipal hotlines exist for reporting drainage or service problems. In practice, information is scattered across websites, social media posts, formal notices, and office counters.

One evening, heavy rain begins before her children return from tuition class. Amara receives a message about possible flooding and road restrictions, but the wording is formal and does not clearly explain whether her street is affected, which bus route is safe, or whether she should move important items to higher shelves.

Her Tamil-speaking neighbour asks for help understanding a municipal waste notice. Another older resident wants to report a blocked drain but does not know which office to contact. Amara tries searching online, but different pages use different terms for the same place, and the instructions are not written in simple language.

At the same time, a community meeting is announced about drainage improvements and public-space planning. Amara wants to attend, but the notice is lengthy and technical. She is unsure whether residents can raise concerns, whether translation will be available, or what documents they need to bring.

A turning point comes when the local community centre pilots VosynCore as a municipal communication access tool. Amara can paste a flood alert, upload a photo of a notice, or ask a question by voice in Sinhala, Tamil, or English. The platform explains the message in simple terms and identifies the next action.

She hears a clear voice explaining that her area is under a localized flood watch. VosynCore lists practical household steps, summarizes safe travel advice, and provides the correct reporting route for blocked drains. The guidance is not a generic translation; it is organized around what she needs to do next.

Later, Amara uses the same tool to understand the community consultation notice. VosynCore summarizes the meeting purpose, explains that residents can share drainage and mobility concerns, and helps her prepare a short question about repeated waterlogging near the bus stop.

This narrative underscores why VosynCore is relevant to SDG 11. Amara’s story is fictional, but it reflects a real urban communication gap: sustainable cities require understandable municipal information as much as physical infrastructure.

By replacing complex, fragmented city communication with localized and interactive voice guidance, VosynCore helps bridge the final mile between municipal systems and the residents who rely on them.

Solution Overview: VosynCore's AI-Powered Municipal Communication Layer

VosynCore is an AI-driven platform designed to bridge language gaps in real time, enabling residents to understand municipal notices, emergency alerts, transport updates, waste instructions, and urban planning information through voice-enabled, multilingual support.

This core capability directly addresses the barriers that prevent residents from acting on available city information. Instead of requiring users to interpret formal documents, technical instructions, or fragmented alerts alone, VosynCore transforms information into practical guidance.

Key Innovations of VosynCore:

- **Real-Time Municipal Alert Translation: VosynCore can convert flood warnings, road closures, transport disruptions, public-health city notices, and service interruptions into Sinhala, Tamil, and English voice-based explanations. The emphasis is on action: what happened, who is affected, what to do, and where to get verified support.**
    - **Example: When a municipal authority issues a flood warning, VosynCore can explain the affected area, immediate household actions, safe-route advice, and reporting options in plain language.**
- **Two-Way Conversational Support: City service information is not a one-way broadcast. Residents often need to ask follow-up questions about routes, schedules, addresses, service categories, documents, or reporting steps. VosynCore can guide users through these questions without replacing official channels.**
    - **Example: If a resident wants to report a blocked drain, the user can describe the issue by voice and VosynCore can help identify the service category, required location details, and the likely municipal reporting channel.**
- **Notice-to-Voice Document Accessibility: Many municipal messages appear as PDFs, posters, scanned notices, or social media images. VosynCore can support OCR-linked extraction, translation, and voice reading for municipal documents.**
    - **Example: If a resident receives a waste-collection poster or a community consultation flyer, the platform can summarize what it says, list dates and locations, and explain what the resident needs to do.**
- **Participatory Planning Communication: Sustainable cities require resident participation. VosynCore can simplify planning notices, consultation materials, mobility plans, and public-space proposals so more people can engage meaningfully.**
    - **Example: Before a town meeting, residents can receive a plain-language explanation of a proposed drainage or transport project and prepare questions in Sinhala, Tamil, or English.**
- **Data privacy and Safety: Municipal communication can involve addresses, emergency risks, and vulnerable residents. VosynCore should include privacy protection, verified source labels, and escalation prompts for urgent situations.**

Together, these features make VosynCore a practical urban resilience tool. It does not replace municipal authorities, transport agencies, waste services, or disaster-management officials; it helps residents understand and use those systems more effectively.

Alignment with SDG Targets & Global Frameworks

Through its context-aware AI translation platform, VosynCore addresses communication barriers that prevent residents from benefiting fully from urban services, municipal planning, transport systems, disaster-risk reduction, and environmental management.

SDG 11 Targets Addressed:

Target 11.2: Provide access to safe, affordable, accessible and sustainable transport systems for all

Urban mobility depends on more than roads and vehicles. Residents need understandable route updates, road-closure alerts, safety messages, and accessibility information. Research on urban mobility in Sri Lanka emphasizes human-centred, context-sensitive planning rather than infrastructure-only approaches.

- **VosynCore's Role: By translating and simplifying transport guidance, mobility notices, and route disruption information, the platform helps residents make safer and more informed travel decisions.**

Target 11.3: Enhance inclusive and sustainable urbanization and participatory planning

Inclusive planning requires residents to understand proposals, meeting notices, service changes, and feedback channels. Technical planning language can discourage participation, especially among older adults, low-literacy users, and multilingual communities.

- **VosynCore's Role: The platform can convert planning documents and consultation notices into plain-language voice explanations, helping residents participate in neighbourhood-level decision-making.**

Target 11.5: Reduce disaster impacts and losses caused by disasters

Urban floods create immediate risks when residents do not understand warnings, evacuation instructions, blocked-route information, or reporting mechanisms. Colombo research shows that urban expansion, drainage inadequacy, and reduced water-retention areas increase flood vulnerability.

- **VosynCore's Role: By delivering multilingual flood alerts, preparedness steps, and reporting guidance, VosynCore can support earlier household action and better community response during urban flood events.**

Alignment with Impact Measurement Frameworks

Vosyn is deeply committed to tracking, auditing, and reporting its Social Return on Investment (SROI) using measurable indicators that link platform activity to resident comprehension, service task completion, emergency readiness, and community participation. For SDG 11, success should be measured through practical urban outcomes such as alert comprehension, issue reporting quality, transport-update understanding, waste-instruction compliance, and consultation participation.

For more information on how we meet compliance frameworks, visit our Compliance Document.

KPI Mapping & Other Frameworks

We have identified specific KPIs to measure VosynCore's performance and impact. The table below shows how each indicator aligns with SDG 11 targets, UN indicator logic, IRIS+ impact measurement, and broader ESG or municipal resilience reporting frameworks.

Key Performance Indicator (KPI)	SDG 11 Target & Indicator	Aligned IRIS+ Metric	Other Framework Alignment
Residents Reached with Municipal Guidance: Number of residents receiving VosynCore-supported flood, transport, waste, or municipal notice explanations, disaggregated by language, age, gender, location, and flood exposure where appropriate.	**SDG 11.3 Inclusive and sustainable urbanization; SDG 11.b resilience planning. Tracks access to understandable city-service communication.**	**Client Individuals: Total number of individuals served through community and municipal communication support.**	**GRI 413-1 (Local community engagement): Demonstrates community participation and local service access; municipal resilience reporting.**
Flood Alert Comprehension Rate: Percentage of users correctly understanding flood warnings, safe-route guidance, household preparedness steps, and reporting instructions after using VosynCore.	**SDG 11.5 Reduce disaster impacts and losses; SDG 11.b integrated disaster risk reduction and resilience.**	**Outcome: Improved preparedness and risk understanding among clients or beneficiaries.**	**Sendai Framework alignment: Supports risk communication, preparedness, and community-level disaster response.**
Transport and Service Update Understanding: Number and percentage of residents correctly interpreting road closures, public transport disruptions, utility interruptions,	**SDG 11.2 Safe, affordable, accessible and sustainable transport systems for all.**	***Product/Service Accessibility: Number of users accessing transport or service information in a usable format.***	**GRI 203-2 (Significant indirect economic impacts): Supports access to essential urban services and mobility.**

and municipal service changes.			
Municipal Issue Reporting Completion: Percentage of users who successfully identify the correct channel and complete reports for blocked drains, waste issues, damaged public spaces, or unsafe roads.	**SDG 11.3 Participatory urban planning; SDG 11.6 reduced environmental impact of cities through improved local service response.**	***Client Outcome: Improved access to services; user task completion and service navigation indicators.***	**GRI 413-1 and public-sector service-quality reporting: Measures practical community benefit and feedback loops.**
Waste Communication Engagement: Number of waste-sorting, recycling, collection schedule, and public awareness interactions completed in Sinhala, Tamil, and English.	**SDG 11.6 Reduce adverse environmental impact of cities, including municipal waste management.**	**Environmental Impact: Clients supported with sustainable behaviour guidance and resource-efficiency information.**	**GRI 306 (Waste): Supports waste-awareness, source segregation, recycling, and community participation.**
Public Consultation Participation Support: Number of users accessing simplified planning notices, meeting summaries, and feedback guidance for urban mobility, drainage, waste, or public-space projects.	**SDG 11.3 Inclusive and participatory urban planning; SDG 11.b resilience planning.**	**Stakeholder Engagement: Number of community members supported to understand and participate in planning processes.**	**GRI 413-1 and urban governance reporting: Demonstrates inclusion in consultation and community engagement.**

Table 1: VosynCore SDG 11 Performance Matrix, including KPI mapping, UN indicator harmonization, and municipal resilience reporting alignment.

By establishing this standardized, transparent KPI framework, VosynCore bridges the operational gap between multilingual communication support and measurable urban resilience outcomes.

For municipalities, disaster-management agencies, transport authorities, community organizations, and impact investors, this framework supports evidence-based monitoring of whether residents are receiving and acting on critical urban information.

For more information on the compliance frameworks that govern our operations, visit our Compliance Document.

Impact Pathway

VosynCore's Theory of Change (TOC) for SDG use cases outlines how inputs and activities translate into outputs, outcomes, and long-term impact. For SDG 11, the pathway begins with localized municipal content, verified alerts, partner workflows, and multilingual voice interfaces. These inputs enable residents to understand city notices, report local issues, respond to flood warnings, follow transport updates, and participate in planning processes.

Learning Agenda, Pilot Roadmap & Scalability Strategy

Sri Lanka Inclusive Urban Resilience Pilot

VosynCore's SDG 11 initiative is not a one-off deployment; it is conceived as a learning journey that tests how multilingual municipal communication can improve resident safety, service access, and participation in Sri Lankan cities.

Learning Agenda

We have formulated key learning questions to guide our SDG 11 pilot and inform future scaling. These questions focus on resident comprehension, flood response, waste-management behaviour, transport information access, and municipal reporting confidence.

1. Effectiveness in Municipal Alert Comprehension: How much does multilingual voice guidance improve residents understanding of flood warnings, road closures, waste notices, and service disruptions?

- **Goal: Compare comprehension, recall, and correct-action rates between residents receiving standard notices and residents receiving VosynCore-supported explanations.**
- **Example: Is there a measurable increase in residents who correctly understand when to avoid a flooded road, where to report a blocked drain, or how to follow a revised waste-collection schedule?**

2. Improvement in Local Issue Reporting: Does VosynCore help residents report urban service problems more accurately and confidently?

- **Goal: Measure changes in user confidence, completeness of reports, time to identify the correct authority, and the number of reports that contain usable location and problem details.**

- **Example: Do residents submit clearer reports about drainage blockages, missed collection, unsafe roads, damaged public spaces, or transport disruptions after using voice-guided support?**

3. Participation in Community Planning: Can VosynCore increase meaningful participation in local planning consultations and neighbourhood improvement discussions?

- **Goal: Evaluate whether simplified planning notices and multilingual explanations improve attendance, question preparation, feedback submission, and resident trust.**

- **Example: Do residents better understand proposed drainage, mobility, or public-space projects and feel more confident raising neighbourhood concerns?**

To answer these questions, our Monitoring, Evaluation, and Learning (MEL) team will use a mixed-methods approach: pre/post comprehension checks, usage analytics, focus groups, resident diaries, municipal staff interviews, and comparison of report quality before and after deployment.

For more information, please visit our “Impact Pathway” document.

Pilot Roadmap

The pilot will proceed in iterative stages to ensure that VosynCore is safely and effectively integrated with municipal communication, disaster preparedness, community centres, and resident feedback channels.

Months 0-3 (Preparation and Co-Design): Establish Partnerships and Foundations

- Finalize agreements with municipal partners, disaster-management contacts, community centres, transport information providers, and waste-management stakeholders.

- Conduct co-design workshops with residents from flood-prone neighbourhoods, older adults, women, Tamil-speaking and Sinhala-speaking users, youth, and frontline municipal staff to identify priority communication pain points.

- Prepare localized Sinhala, Tamil, and English content libraries for flood preparedness, drainage reporting, transport disruptions, waste sorting, public-space notices, and community consultation processes.

- **Milestones: Secure partner approvals; configure the initial VosynCore municipal communication workflows; validate terminology; prepare resident onboarding materials and safeguarding protocols.**

Months 4-9 (Implementation and Monitoring): Field Deployment and Feedback Loops

- Roll out VosynCore-supported access points in selected community centres, municipal counters, resident associations, and mobile/browser channels for neighbourhood communication support.
- Implement real-time monitoring of translation accuracy, user comprehension, report completion, alert understanding, and resident feedback while maintaining privacy and verified-source safeguards.
- Facilitate monthly learning circles with residents, municipal staff, and community volunteers to refine terminology, examples, escalation prompts, and service categories.
- **Milestones: Achieve target engagement levels within the pilot cohort; collect interim quantitative and qualitative evidence; update the model based on local usage and resident feedback.**

Months 10-12 (Evaluation and Scaling Prep): Impact Assessment and Synthesis

- Execute endline surveys, usage analysis, report-quality reviews, flood-alert comprehension checks, and final focus groups to measure pilot outcomes.
- Synthesize findings into an Inclusive Urban Resilience and Municipal Communication report for city authorities, community organizations, disaster-risk partners, and potential investors.
- Refine AI models, content libraries, OCR workflows, and escalation rules based on pilot evidence to prepare for broader municipal deployment.
- **Milestones: Publish a validated SDG 11 impact report; finalize the roadmap for Phase 2 expansion into additional urban councils, flood-prone communities, and municipal service domains.**

Scalability Strategy

Looking beyond the pilot, our ambition is to scale vertically by embedding VosynCore within municipal communication channels and horizontally by adapting the same model to other Sri Lankan cities and urban service domains.

- **Municipal Integration: We aim to embed VosynCore into existing local authority websites, emergency alert systems, service counters, hotlines, and community centres.**
    - **Example: Following a successful pilot, a municipal council could add VosynCore as a voice-guidance layer for flood notices, waste schedules, and service reporting workflows.**

- **Urban-Service Replicability: The modular architecture allows new city-service domains, neighbourhood names, and local instructions to be added without rebuilding the platform from scratch.**
    - **Example: Once the model is stabilized for flood alerts and waste guidance, the same framework can support public transport changes, roadworks notices, park maintenance updates, and consultation materials.**
- **API-First Partnership Model: To achieve exponential growth, we will offer VosynCore as an API for municipal portals, disaster-alert dashboards, transport apps, and community information platforms.**
- **Example: A city website, transport-information provider, or public-service kiosk could integrate VosynCore's translation and voice explanation layer into its existing communication workflow.**
- **Community-Led Expansion: We leverage a train-the-trainer model where successful pilot facilitators become champions for broader neighbourhood adoption.**
    - **Example: Community officers, resident association leaders, youth volunteers, women's groups, and disaster-preparedness volunteers can support onboarding and feedback collection.**

This strategy ensures that VosynCore evolves from a localized municipal communication aid into a scalable standard for inclusive urban resilience. It strengthens existing city systems by making them easier to understand, use, and trust.

Modelled Projections

As VosynCore moves toward field deployment, we have developed a series of evidence-informed projections based on the expected relationship between accessible municipal communication, resident comprehension, timely reporting, and safer urban behaviour.

Projected Outcomes (Short to Medium Term):

In the initial 12-24 months of deployment, our focus is on establishing a baseline of municipal communication comprehension, emergency readiness, issue reporting, and participation in neighbourhood planning.

- **Improved Flood Alert Understanding: A 35%-45% projected increase in residents correctly understanding flood warnings, safe-route guidance, and household preparedness steps when messages are delivered through localized voice support.**
- **Higher Municipal Task Completion: Up to 50% projected improvement in residents ability to complete selected tasks such as reporting blocked drains, following waste-sorting instructions, or identifying the correct service channel.**

- **Better Transport and Service Disruption Awareness: A measurable increase in residents ability to understand route changes, road closures, collection delays, and local service updates without relying only on informal networks.**

- **Improved Public Consultation Participation: Increased attendance, feedback submission, and question preparation for selected neighbourhood planning or resilience consultations supported by simplified multilingual explanations.**

Projected Outcomes (Medium to Long Term):

Looking toward 2030, the integration of VosynCore into municipal communication and disaster-resilience systems could support broader city-level improvements in resident safety, trust, service use, and participatory planning.

- **Resilient Community Communication: Flood-prone and service-vulnerable neighbourhoods receive clearer, faster, and more actionable guidance before, during, and after urban disruptions.**

- **Strengthened Municipal Service Responsiveness: Local authorities receive clearer resident reports, better categorized service requests, and more feedback from communities that may otherwise remain underrepresented.**

- **Cleaner and More Participatory Urban Environments: Waste-management and public-space instructions become easier to understand, supporting better sorting behaviour, reporting, and community engagement.**

- **Scalable Model for Inclusive Cities: VosynCore can become a reusable approach for making municipal communication accessible across transport, drainage, waste, public-space, and emergency-preparedness domains.**

Conclusion: High-Potential Innovation for SDG 11

VosynCore fuses innovative AI with human-centred urban design to address one of the most practical barriers to sustainable cities: the gap between municipal systems and resident understanding. In Sri Lanka, this gap can affect flood preparedness, waste collection, transport access, service reporting, and participation in local planning.

This initiative demonstrates how inclusive AI amplifies the impact of existing city systems. It does not build drains, roads, buses, or waste facilities; rather, it helps residents understand how to use, report, and respond to those systems more effectively.

Key Takeaways:

- **Systemic Force Multiplier: VosynCore does not replace municipal services; it unlocks them. By reducing language and comprehension barriers, it improves the usefulness of existing alerts, notices, hotlines, and service workflows.**

- **Measurable Social Return on Investment (SROI): The platform converts communication access into practical outcomes such as faster flood response, clearer reports, better waste compliance, improved transport awareness, and stronger participation.**

- **Distinct SDG 11 Focus: This use case is intentionally framed around city-level resilience and municipal communication. It avoids overlap with SDG 6 water safety, SDG 7 energy, SDG 8 employment, SDG 9 digital infrastructure, and SDG 10 marginalized-service inequality.**

- **Rigorous Institutional Alignment: Built for accountability, the platform's roadmap aligns with SDG Targets 11.2, 11.3, 11.5, 11.6, and 11.b, while connecting to municipal resilience, ESG, and community-impact reporting.**

Achieving SDG 11 requires more than construction and infrastructure investment. It requires residents to receive clear, trusted, and timely information that helps them move safely, prepare for hazards, manage waste responsibly, and participate in decisions about their neighbourhoods.

The SDG 11 initiative offers a high-potential innovation that demonstrates the promise of AI for sustainable cities. By making municipal communication accessible in Sinhala, Tamil, and English, VosynCore can help build safer, more resilient, and more participatory urban communities in Sri Lanka.

References

Institutional References (UN & Global Reports)

United Nations Department of Economic and Social Affairs. (2025). Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable. https://sdgs.un.org/goals/goal11

United Nations Statistics Division. (2025). The sustainable development goals report 2025: Goal 11. https://unstats.un.org/sdgs/report/2025/

UN-Habitat. (2020). World cities report 2020: The value of sustainable urbanization. UN-Habitat. https://unhabitat.org/world-cities-report-2020

UNDRR. (2015). Sendai Framework for Disaster Risk Reduction 2015-2030. United Nations Office for Disaster Risk Reduction. https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030

World Bank. (2023). Sri Lanka development update 2023. World Bank Group. https://www.worldbank.org/en/country/srilanka/publication/sri-lanka-development-update

Asian Development Bank. (2021). Sri Lanka urban development and infrastructure sector materials. Asian Development Bank. https://www.adb.org/countries/sri-lanka/main

Scientific & Academic References

Jayalath, J. A. C. D., & Kulatunga, U. (2025). Integrating socio-cultural and environmental dimensions into urban mobility planning: A case study of Kiribathgoda Town, Sri Lanka. Bhumi, 12(2), 35-49. https://doi.org/10.4038/bhumi.v12i2.146

Abeysinghe, U., Hewawasam, H. U. C. P., & Manatunga, J. (2025). Disaster-resilient cities: Temporal analysis of urban land use land cover changes and urban floods from 1999 to 2022: Case study of Colombo, Sri Lanka. Proceedings of the 13th World Construction Symposium, 598-609. https://doi.org/10.31705/WCS.2025.45

Alahakoon, L., Cerneckiene, J., Jansons, L., & Tekshan, W. M. M. (2025). A new approach to infrastructure development in Sri Lanka's urban areas. Baltic Journal of Real Estate Economics and Construction Management, 13, 68-79. https://doi.org/10.2478/bjreecm-2025-0006

Tennakoon, S. M. A. V. S., Udayantha, U. L. I., & Wattegama, W. G. E. J. (2025). Waste management research in Sri Lanka: Exploring trends, identifying gaps, and gaining key insights through bibliometric analysis. Management Dynamics in the Knowledge Economy, 13(1), 20-37. https://doi.org/10.2478/mdke-2025-0002

Additional urban resilience literature on flood risk, transport planning, municipal solid waste management, climate adaptation, green infrastructure, participatory urban governance, local government service delivery, and community disaster response

should be integrated as the pilot moves from concept design to municipal field validation.

Pilot Evidence Validation Notes

The SDG 11 pilot should treat all projections as working assumptions until they are tested through field evidence. Baseline and endline data should be collected on resident comprehension, alert understanding, transport-update use, waste-instruction compliance, reporting confidence, and participation in community planning processes.

Municipal communication should be evaluated as a practical urban resilience outcome rather than only a translation outcome. The key question is whether residents understand what to do, where to go, what to report, and how to participate after receiving VosynCore-supported guidance.

Flood-related use cases should include strict safeguards. VosynCore should prioritize verified official alerts, avoid replacing emergency authorities, and guide users toward appropriate hotlines, evacuation instructions, or local responders where urgent risk is identified.

Waste-management and transport use cases should be tested with diverse residents, including older adults, commuters, low-literacy users, Tamil-speaking and Sinhala-speaking communities, apartment residents, informal settlement residents, and small business owners affected by service changes.

Urban resilience monitoring should also consider whether the same communication support works differently across neighbourhoods. Evaluation should disaggregate results by language, gender, age, location, digital access, and flood exposure so that the pilot can identify who benefits and where additional support is needed.